

Unified Rectification Operators

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This document defines both rectification methodologies:

1. Limit Rectification Operator (LRO)
2. Trigonometrical Rectification Operator (TRO)

Each operator modifies a function locally while preserving global structure, but they achieve rectification through different mechanisms.

1. Limit Rectification Operator (LRO)

Definition

Let $f(x)$ be a differentiable real-valued function.

A Limit Rectification Operator uses limiting behavior to produce a plateau region where the slope tends to zero without requiring an explicit piecewise definition.

Core concept:

As $x \rightarrow p$, $f'(x) \rightarrow 0$

Construction

A typical LRO form:

$(L f)(x) = f(x)$ for $x < p$

$(L f)(x) = f(p) + \varepsilon \sin(k(x - p))$ for $x \geq p$

with:

$\varepsilon \rightarrow$ very small amplitude

$k \rightarrow$ very low frequency

The derivative is never exactly zero, but visually and numerically the plateau is flat.

Properties

- Produces a soft, asymptotic plateau

- Derivative tends to zero
- No explicit piecewise carving needed
- Smooth and continuous if f is smooth
- Can be made C^∞ by smoothing the transition
- Retains global equivalence of $f(x)$
- The plateau is a limit, not a replacement

Interpretation

The LRO “rectifies” by bending the derivative until it becomes effectively zero. It does not modify the original function directly, only its behavior.

Applications

- Soft controller flattening
 - Smooth saturation
 - Invisible derivative damping
 - Asymptotic stabilization
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2. Trigonometrical Rectification Operator (TRO)

Definition

The TRO replaces a region $[p, q]$ of the function with a dual–trigonometric, zero-boundary, normalized oscillatory core, preserving global invariants.

Window Function

$w(x) = 1$ on $[p, q]$

$w(x) = 0$ outside

C^∞ transitions recommended

Dual■trig core:

$h(t) = \sin(\pi t) + \alpha \sin(2\pi t)$

$t = (x - p)/(q - p)$

Rectified core:

$g(x) = n + A \cdot h((x - p)/(q - p))$

Operator definition:

$$(Tf)(x) = (1 - w(x)) f(x) + w(x) g(x)$$

Normalization

To preserve global average on $[p, q]$:

$$\int_{-1}^1 h(t) dt = 0$$

or

$$A \int_{-1}^1 h(t) dt = 0$$

Properties

- Dual-trigonometric insertion
- Perfect endpoint matching at $x = p$ and $x = q$
- Global invariants preserved (mean, endpoints, energy)
- Local, reversible deformation
- C^∞ smoothness if w and f are C^∞

Interpretation

The TRO performs a structured replacement of a linear/flat segment with a controlled oscillatory geometry while keeping all global behavior intact.

Applications

- Activation engineering
- Waveform shaping
- Functional smoothing
- Controlled nonlinearity injection
- Rectified plateaus with preserved integrals

3. Relationship Between LRO and TRO

Both operators create rectification regions, but:

- LRO uses limiting derivative collapse
- TRO uses explicit dual-trigonometric replacement

Both:

- preserve global behavior
- allow an advanced system to treat them as effectively linear
- introduce controlled, localized nonlinearity
- remain smooth and differentiable
- are injective on their modification regions

4. Unified Rectification Framework

Both operators belong to a larger class:

Rectification Operators ■

Where each element ■ modifies f only on a compact interval while preserving:

- endpoint continuity
- global normalization
- smoothness
- functional identity outside the region

Examples:

LRO $\in$ ■ (limit-driven)

TRO $\in$ ■ (normalization-driven)

5. Summary

LRO:

“Flatten by limiting behavior”

Plateau emerges asymptotically

TRO:

“Flatten by structured replacement”

Plateau contains a normalized dual-trig core

Both are mathematically valid, smooth, localized, and globally neutral rectification mechanisms.